

$\tau = R \cdot C \rightarrow$ time constant

$$= 100 \cdot 5 \cdot 10^{-9} = \underline{\underline{5 \cdot 10^{-7} \text{ F} \cdot \Omega = 5 \cdot 10^{-7} \text{ s}}}$$

(1)

* τ is the value at which the V_{os} reaches its 63% of its peak value during rise time.

rising edge: $\rightarrow V(t) = 10 \cdot [1 - e^{-t/5 \cdot 10^{-7}}]$
 falling edge: $\rightarrow V(t) = 10 \cdot e^{-t/5 \cdot 10^{-7}}$

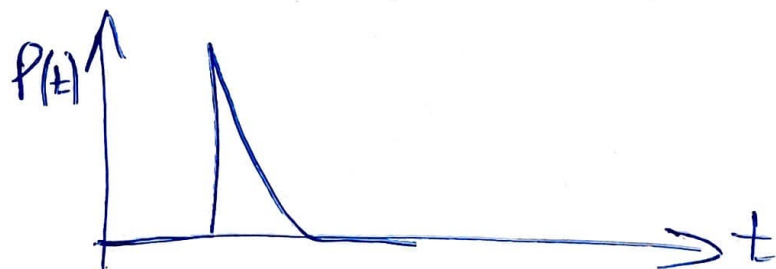
* for the low-side switch; the low-side switch discharges the capacitor, whereas the high-side switch charges the capacitor.

Low-side switch: $V(t) = 10 \cdot e^{-\frac{t}{5 \cdot 10^{-7}}}$
 $i(t) = C \frac{dV}{dt} = 5 \cdot 10^{-9} \cdot \frac{1}{5 \cdot 10^{-7}} \cdot 10 \cdot e^{-\frac{t}{5 \cdot 10^{-7}}}$

$$i(t) = 0.1 \cdot e^{-\frac{t}{5 \cdot 10^{-7}}}$$

Switching Loss:

$$P_{\text{loss}} = V \cdot I = 10 \cdot e^{-\frac{t}{5 \cdot 10^{-7}}} - 0.1 \cdot e^{-\frac{t}{5 \cdot 10^{-7}}} = e^{-\frac{2t}{5 \cdot 10^{-7}}}$$



Average Power:

(2)

$$P_{avg} = \frac{1}{10 \cdot 10^{-6}} \int_0^{10 \cdot 10^{-6}} e^{-\frac{2t}{5 \cdot 10^{-7}}} dt$$

$$f = 100 \text{ kHz} \rightarrow \underline{T = 10 \mu\text{s}}$$

$$P_{avg} = \frac{1}{10 \times 10^{-6}} \int_0^{10 \times 10^{-6}} e^{-\frac{2t}{5 \cdot 10^{-7}}} dt \rightarrow a = \frac{2}{5 \cdot 10^{-7}} = 4 \cdot 10^6$$

$$\frac{1}{10^{-5}} \int_0^{10^{-5}} e^{-4 \cdot 10^6 t} dt \Rightarrow \left[-\frac{1}{4 \cdot 10^6} e^{-4 \cdot 10^6 t} \right]_0^{10^{-5}} \cdot \frac{1}{10^{-5}}$$

$$\underline{\text{Upper Limit:}} \rightarrow t = 10^{-5}$$

$$-\frac{1}{4 \cdot 10^6} e^{-4 \cdot 10^6 \cdot 10^{-5}} = -\frac{1}{4 \cdot 10^6} e^{-40} \approx \underline{0}$$

$$\underline{\text{Lower Limit:}} \rightarrow t = 0$$

$$-\frac{1}{4 \cdot 10^6} e^0 = -\frac{1}{4 \cdot 10^6}$$

$$= \frac{1}{10^{-5}} [\text{Upper Limit} - \text{Lower Limit}]$$

$$= 10^5 [0 + 0.25 \cdot 10^{-6}]$$

$$= 10^5 [0.25 \cdot 10^{-6}]$$

Low-side

Switch $\rightarrow$

$$\boxed{P_{avg} = 25 \text{ mW}}$$

$$P_{LS} = P_{HS} \quad \text{Total Half-Bridge Loss} = 2 \cdot 25 = \underline{50 \text{ mW}}$$